

## The $(\log 2, \log 3]$ step:

Connes–Consani’s archimedean operator rebuilt, the semi-local  $\{\infty, 2\}$  Sonin space on a slice,

and why the compact-remainder mechanism does not cross the place 2

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3 September 2026 — consolidates notebook §80–85 of `riemann-weil-phenomenology`

### Abstract

Weil positivity is known, without RH, for test functions supported in  $[2^{-1/2}, 2^{1/2}]$  (window length  $L = \log 2$ ); the proof of Connes and Consani (Selecta 2021) writes the archimedean Weil functional as a positive trace on Sonin’s space minus a remainder whose conditioned operator is  $-2\epsilon'(1^+)\text{Id}$  plus a compact part with a single eigenvalue above 1. Crossing the threshold of the prime 2 — positivity for  $L \in (\log 2, \log 3]$  — is the first open step of their semi-local program. We (i) rebuild their archimedean operator and reproduce every published digit; (ii) measure that its compact part acquires a second eigenvalue above 1 at  $L \approx 1.01$ , while the archimedean form itself stays positive on the conditioned hyperplane at  $L = \log 3$ ; (iii) show that the prime-2 term cannot be added to the compact remainder; (iv) build the semi-local  $\{\infty, 2\}$  Sonin space on the  $\text{ord}_2 = 0$  slice, where the adelic Fourier transform has the closed form  $\mathfrak{F}g(\rho) = \frac{1}{2}[\sum_{n \geq 0} \hat{g}(2^n \rho) - \hat{g}(\rho/2)]$ , and find that the compression is not trace class, that the remainder is log-divergent at  $\rho = 1$  and spiked at  $\rho = 2$ , and that the conditioned remainder is *essentially positive* — the sign that made the archimedean argument work is reversed; (v) check the construction against the semi-local trace formula of Connes (1999): the archimedean distribution is reproduced to 1–4% and the 2-adic place appears exactly at  $\lambda = 2^{\pm 1}$ , its weights not yet converged at  $\Lambda \leq 8$ . Everything is measurement and calibration; the conclusion is that the  $(\log 2, \log 3]$  step requires a mechanism of a different nature, and the construction says which term is in the way. Nothing here bears on RH.

## 1 The archimedean mechanism and where it stops

Connes–Consani prove, for  $g$  supported in  $[2^{-1/2}, 2^{1/2}]$  with  $\hat{g}(\pm i/2) = \hat{g}(0) = 0$ , that  $W_\infty(g * g^*) \geq \text{Tr}(\vartheta(g)S\vartheta(g)^*) \geq 0$ . The engine is the identity (their Theorem 4.7)  $\text{Tr}(\vartheta(f)S) = W_\infty(f) + E(f)$ ,  $S$  the projection on Sonin’s space  $\mathcal{S}(1, 1)$  (even functions vanishing with their Fourier transform on  $[-1, 1]$ ),  $E(f) = \int f(\rho^{-1})\epsilon(\rho) d^*\rho$  with  $\epsilon$  explicit in the prolate spheroidal functions of bandwidth  $2\pi$ . After the vanishing conditions are imposed through  $Q = -(\rho\partial_\rho)^2 + \frac{1}{4}$ , the remainder becomes  $E \circ Q(\xi * \xi^*) = \langle \xi | N_I \xi \rangle$ ,  $N_I = -2\epsilon'(1^+)(\text{Id} - K_I)$ ,  $K_I$  compact of Hilbert–Schmidt class on  $L^2(I)$ ,  $I = [-\frac{1}{2}\log 2, \frac{1}{2}\log 2]$ . Positivity follows if  $K_I \leq 1$ ; they compute that  $K_I$  has exactly one eigenvalue above 1 (1.0516), neutralized by one linear condition, with constant  $c \in (13, 17)$ . The kink of  $\epsilon$  at  $\rho = 1$  ( $\epsilon'(1^+) = 22.9965$ ) is what produces the  $-2\epsilon'\text{Id}$ ; the finiteness  $\delta(1) = \sum \lambda(n)^2 = 2.2375$  is what makes the remainder a regular kernel. The whole argument stops at  $L = \log 2$ , to one eigenvalue.

## 2 Reconstruction and calibration

`code/cc_arch.py` rebuilds the objects: prolates  $PS_{2n,0}(2\pi, \cdot)$ , eigenvalues  $\lambda(n)$  of the finite Fourier transform,  $\xi_n$ , the analytic continuation  $\xi_n^{an}(x) = \lambda(n)^{-1} \int_{-1}^1 \xi_n(t) \cos(2\pi tx) dt$ ,  $Q\epsilon$  from their formula (99), and  $K_I$  by their  $q \rightarrow 1$  Toeplitz discretization ( $\omega = 2 \times 10^{-3}$ ).

| quantity                                                | rebuilt                                  | Connes–Consani                                   |
|---------------------------------------------------------|------------------------------------------|--------------------------------------------------|
| $\lambda(0), \lambda(1), \lambda(2), \lambda(3)$        | 0.999971, -0.979485, 0.524086, -0.058977 | 0.999971, -0.979485, 0.524086, -0.0589766        |
| $t(0), t(1), t(2), t(3)$                                | 11.9719, 8.77574, 2.20528, 0.0434        | 11.9719, 8.77574, 2.20528, 0.0433983             |
| $\epsilon'(1^+)$                                        | 22.9965                                  | 22.9965                                          |
| $\lambda_{\max}(K_I), \lambda_2, \lambda_3$ at $\log 2$ | 1.05176, 0.68791, 0.0297                 | 1.05177 ( $\omega = 10^{-3}$ ), 0.687925, 0.0289 |

### 3 Beyond $\log 2$ in the archimedean frame

The second eigenvalue of  $K_I$  crosses 1 at  $L \approx 1.01$  ( $\mu \approx 2.74$ ): one eigenvalue above 1 on  $(\log 2, 1.01)$ , two on  $(1.01, \log 3]$  (1.0899, 1.0395 at  $\log 3$ ). Yet the archimedean form itself is still positive in their frame: on the Galerkin space  $V_{31}$  at  $\mu = 3$ , the archimedean form with the pole removed has one negative direction on  $V$  (along the pole) and *none* on the hyperplane  $c(f) = 0$  of the vanishing condition. Their sufficient condition  $K_I \leq 1$  thus already fails at  $\log 3$  while the conclusion holds — the Sonin trace compensates both directions of the remainder.

The prime-2 term cannot be added to the remainder. On the conditioned functions  $\xi$ ,  $W_2 \circ Q(\xi * \xi^*) = \frac{\log 2}{\sqrt{2}} [\frac{1}{4}F(\log 2) - F''(\log 2)]$ ,  $F = \xi * \xi^*$ , is a differential form, unbounded on  $L^2(I)$ ; on the side  $g = (\frac{1}{2} - \partial)\xi$  it is a bounded translation form, but the archimedean identity term becomes  $-2\epsilon' \|Y * g\|^2$ , a weak norm that does not control  $G(\log 2)$  for oscillating  $g$ . In either presentation  $E + W_2 \leq 0$  is false, although  $W_\infty - W_2 \geq 0$  holds numerically at  $\log 3$  ( $\lambda_{\min} = 5.9 \times 10^{-8}$  on  $V_{31}$ ): the positivity comes from the positive trace absorbing  $W_2$ , not from a bound on the remainder. This is the operator form of a Galerkin measurement: at  $L = \log 3$  the archimedean form  $Q_\infty = \text{pole} + \text{arch}$  has exactly one negative direction  $v$  (eigenvalue  $-0.074$ ),  $-T_2$  is positive on it with margin ( $Q(v) = +0.050$ , using 5% of  $T_2$ 's mass), and the razor  $5.9 \times 10^{-8}$  lives in the positive complement.

### 4 The semi-local $\{\infty, 2\}$ Sonin space on a slice

Following Connes (1999, §VII),  $X_S = (\mathbb{R} \times \mathbb{Q}_2)/\pm 2^{\mathbb{Z}}$ ,  $L^2(X_S)$  is the completion of  $\mathcal{S}(\mathbb{A}_S)$  for  $\|f\|^2 = \int_{\mathcal{D}} |\sum_q f(qx)|^2 |x| d^*x$ , and  $F = F_\infty \otimes F_2$  is unitary on it (his Lemma 1). Functions invariant under  $\mathbb{Z}_2^*$  are determined by their restriction  $g$  to the shell  $\text{ord}_2 = 0$ , and on this slice

$$\mathfrak{F}g(\rho) = \frac{1}{2} \left[ \sum_{n \geq 0} \hat{g}(2^n \rho) - \hat{g}(\rho/2) \right], \quad \hat{g}(\xi) = 2 \int_0^\infty g(r) \cos(2\pi r \xi) dr. \quad (1)$$

**Proposition 1.**  $\mathfrak{F}$  is unitary and involutive on  $L^2([0, \infty))$ ; hence self-adjoint.

*Proof.* Write  $\mathfrak{F}g = \sum_{k \geq -1} c_k \hat{g}(2^k \cdot)$  with  $c_{-1} = -\frac{1}{2}$ ,  $c_k = \frac{1}{2}$  ( $k \geq 0$ ). For  $j \neq 0$  the cross coefficient  $\sum_k c_k c_{k+j} 2^{-k}$  vanishes ( $-\frac{1}{2} + \frac{1}{2}$ ), and the diagonal  $\sum_k c_k^2 2^{-k} = \frac{1}{2} + \frac{1}{2} = 1$ :  $\|\mathfrak{F}g\| = \|\hat{g}\| = \|g\|$ . Involutivity is  $F^2 = \text{id}$  on even functions of  $\mathbb{A}_S$ , which descends. Numerically (code/semilocal.py, tests):  $\|\mathfrak{F}g\|^2/\|g\|^2 = 1.015$ ,  $\mathfrak{F}^2 g = 0.94 g$  on the support of a Gaussian bump, the deviations being quadrature.  $\square$

The cutoff  $P_1 = \mathbf{1}_{[0,1]}$  and the scaling  $\vartheta(\lambda)g(r) = \lambda^{-1/2}g(r/\lambda)$  are unchanged; only  $\hat{P}_1 = \mathfrak{F}P_1\mathfrak{F}$  and Sonin's space change. Three facts, all computed with exact cell averaging (a first pointwise assembly aliased the lacunary terms and produced a spurious 43% asymmetry; it is retracted in the notebook):

- *The compression is not trace class.* The eigenvalues of  $P_1 \mathfrak{F} P_1$  decay slowly (0.998, -0.990, 0.613, -0.554, 0.4... and  $\sum \lambda_n^2$  grows with resolution (4.24 at  $N = 200$ , 4.75 at  $N = 400$ ; archimedean: 2.2375, stable). The kernel  $\sum_n \cos(2\pi 2^n pr)$  is not square integrable on  $[0, 1]^2$ .

- *The remainder is log-divergent at  $\rho = 1$  and spiked at  $\rho = 2$ .*  $\delta_S(\rho) = \text{Tr}(\vartheta(\rho^{-1})\hat{P}_1 P_1) = \sum_n \lambda_n \langle \xi_n | \vartheta(\rho^{-1}) \eta_n \rangle$  (CC's Lemma 4.1, which uses only the algebra of two projections). Archimedean control: the closed form  $2\sqrt{\rho} [\text{Si}(2\pi(1+\rho))/(2\pi(1+\rho)) + \text{Si}(2\pi(\rho-1))/(2\pi(\rho-1))]$  is reproduced to 3–4 digits. Semi-local:  $\delta_S \approx -0.65 \ln(\rho-1) + C$  ( $\delta_S(1) = \sum \lambda_n^2 = \infty$ ) — a logarithm where CC had a kink — and a spike at  $\rho = 2$  growing with resolution: the 2-adic Weil term, appearing where it must.
- *The conditioned remainder is essentially positive.* On twenty smooth test functions of  $I$  (sine basis; eigenvalues of  $D \circ Q$  relative to the Gram matrix; `code/dq_sign.py`): archimedean at  $\log 2$ , 2 positive eigenvalues out of 20 (4.77, 1.23) and the rest massed at  $-2.01$  — the  $-2\text{Id}$  of CC's Theorem 3.6 in plain sight, and their Remark 3.9 recovered; at  $\log 3$ , 3 positive. Semi-local at  $\log 2$ : 20 **positive out of 20** (up to 121); at  $\log 3$ , 13 out of 20 (up to 196), the positive part growing with the basis — the  $H^{1/2}$  signature of the logarithm ( $-\ln|x|$  and the finite part of  $1/x^2$  both have positive transforms).

## 5 The verdict

$W_\infty - W_2 = L_S - D_S$  with  $L_S$  a positive trace. Extracting positivity requires  $D_S \circ Q$  essentially negative, as in the archimedean case. It is essentially positive: the positive trace absorbs the place 2 but with a divergent positive surplus that erases the sign of  $W$ . The compact-remainder mechanism of Connes–Consani, transported verbatim to  $\{\infty, 2\}$  at  $\Lambda = 1$ , does not give the  $(\log 2, \log 3]$  step (28th preregistered execution of the repository). The construction also says why: the surplus is the term  $2h(1) \log' \Lambda$  of the semi-local trace formula of Connes (1999), and that theorem states that the finite part of the compressed trace, once this term is subtracted, *is* the Weil functional — no positivity comes for free. The archimedean miracle (a finite  $\delta(1)$  and a kink) belongs to one place.

## 6 Validation against the semi-local trace formula

$\tau_\Lambda(\lambda) = \text{Tr}(\hat{P}_\Lambda P_\Lambda \vartheta(\lambda))$  on the slice (`code/trace_dist.py`,  $\Lambda = 4, 8$ ). Theorem 4 of Connes (1999) predicts  $\tau_\Lambda \rightarrow 2 \log' \Lambda \delta_1 + \tau_\infty + \tau_2$  with  $\tau_\infty = \lambda^{1/2}/2 (1/(1+\lambda) + 1/|1-\lambda|)$  (CC (39)) and  $\tau_2$  point masses at  $\lambda = 2^{\pm m}$ . Archimedean:  $\tau_\Lambda$  matches (39) to 1–4% away from  $\lambda = 1$  (0.7495/0.7529 at 0.4, 2.4986/2.4845 at 0.8, 0.9338/0.9428 at 2.0, 0.6521/0.6495 at 3.0). Semi-local minus archimedean: two sharp peaks exactly at  $\lambda = 2$  and  $\lambda = \frac{1}{2}$  at both  $\Lambda$  — the 2-adic place where the theorem puts it. Their integrated weights (+0.13/−0.08 at  $\Lambda = 4$ , −0.23/−0.35 at  $\Lambda = 8$ , against  $\log 2/\sqrt{2} = 0.49$ ) oscillate and change sign: not converged at  $\Lambda \leq 8$ . The check is conclusive on the location of the places and open on the 2-adic weight; closing it needs  $\Lambda \geq 16$  (matrices  $512 \times 1700$  with forty lacunary Si terms) or an analytic treatment of the 2-adic shells.

## Status and reproducibility

Calibration is exact to the published digits; the semi-local objects are computed with exact cell averaging and covered by fourteen recomputing tests (`tests/test_semilocal*.py`, suite of 128). The sign reversal of the conditioned remainder is a measurement on twenty test functions at two window lengths, not a theorem; the identification of the surplus with the  $\log \Lambda$  term of 1999 is an inference, supported by the log-divergence of  $\delta_S$  at 1. What the note establishes is negative and precise: the first semi-local step of the Connes–Consani program cannot be taken by transporting their  $\Lambda = 1$  Sonin mechanism; it needs a term subtracted before the compact part can be examined. Nothing here bears on RH.